### 7.13.3 Pathogenicity

Source of infection: Contaminated water

Route of entry: Through cuts or abrasions on skin or mucosa

Incubation period: 6–8 days

- ***Leptospira interrogans*** causes a zoonotic disease named Leptospirosis. It is transmitted to humans by direct or indirect contact with water, contaminated by urine of carrier animals (rat and dog).
- Leptospira enter the body through cuts or abrasions on skin or through mucous membranes of the mouth, nose or conjunctiva.
- After an incubation period of 6–8 days. There is onset of febrile (related to fever) illness with Leptospira in blood (Septicemic phase) which lasts for 3–7 days.
- The organisms disappear from the blood and invades liver, kidney, spleen, meninges producing meningeal irritation such as headache, vomiting.
- The pathogen persists in the internal organs and most abundantly in the kidney. Severe Leptospirosis (Weil's disease) is associated with Fever, conjunctivitis (inflammation of conjunctiva), albuminuria (presence of albumin in the urine), jaundice and hemorrhage. It is a fatal illness with hepatorenal (Kidney failure with severe liver damage).

**Clinical manifestations**

- In severe cases, vomiting, headache, irregular fever and intense infection of the eyes.
- Jaundice, Albuminuria (The presence of protein Albumin in the urine) and purpuric hemorrhages sometimes occur on skin and mucosa.
- Aseptic meningitis is common in canicola infection.

### 7.13.4 Laboratory Diagnosis

The diagnosis of Leptospirosis is made by the following ways

- Direct microscopy of blood or urine
- Isolation of pathogen by culture
- Serological tests.

**Direct Microscopy**

**Blood:** Leptospira can be observed in the blood by dark – filed microscope. Blood examination is useful in first week as Leptospira disappear from blood after 8 days.

**Urine:** Leptospira may be present in urine in the 2$^{nd}$ week of the disease and intermittently thereafterup to 6 weeks. Centrifuged deposit of urine may be observed by Dark filed microscopy.

**Culture: B**lood (1$^{st}$ week) and urine (2$^{nd}$–6 week) can be cultured in Korthof's medium. Media are incubated at 37°C for 2 days and then left at room temperature for 2 weeks. Culturesare examined every third day for the presence of Leptospira under DFM.

**Serological tests**

It is very useful method of diagnosis two types of serological tests are used, which are,

a. **Screening tests:** These tests are genus – specific and done using reactive genus



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